

Francisco Cardozo

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Education

PhD, Prevention Science and Community Health. University of Miami, USA - 2025

MSc, Psychology. Universidad de los Andes, Bogotá, Colombia - 2018

BS, Psychology. Universidad Nacional, Bogotá, Colombia - 2014

Trainings

Machine Learning

1. 2025 – Artificial Intelligence and Applications in Prevention Science, SPR Conference, Seattle.
2. 2025 – Introduction of LLMs/AI Workshop. New York Data Science & AI Conference, New York.
3. 2024 – Building Generative AI Applications, PyData Conference, New York.
4. 2024 – R Software in Production, Posit::conf, Seattle.
5. 2023 – Advanced Tidymodels for Machine Learning, Posit::conf, Chicago.
6. 2023 – Advanced Package Development in R, Posit::conf, Chicago.

Statistical Methods

1. 2024 – Network Analysis using TERGM and SIENA, INSA Conference, Edinburgh.
2. 2023 – Advanced Data Analytics Certificate, Google-Coursera.
3. 2023 – Missing Data Using R, Statistical Horizons.
4. 2022 – Research Design for Causal Inference, University of Chicago.
5. 2018 – Multilevel Analysis Applied to Public Health, Universidad de los Andes.
6. 2016 – Statistics in Social Sciences, Universidad de los Andes.

Teaching

1. 2023 – Software Instructor, The Carpentries.
2. 2023 – Teaching Academy Certificate, University of Miami.

Honors

1. 2026 - Nominee, NIH Director's Early Independence Award (DP5), University of Miami.
2. 2025, 2023 – Nominee, Graduate Teaching Assistant Award, University of Miami.
3. 2024 – Scholar Award, PyData Conference. Python.
4. 2024 – Scholar Award, Posit::conf. R software.
5. 2024 – Winner, Data Science Competition Sloboda & Bukowski Cup, SPR.
6. 2020 – Fulbright Scholarship for Doctoral Studies.
7. 2016 – Academic Scholarship, IRENEO ROSSIER, Universidad de los Andes.

Publications

Journal Articles (peer reviewed)

1. Canizares, C., **Cardozo, F.**, Shankar, A., Balise, R., & Morris-Perez, P. (2026). Association between plain-water consumption and suicide attempts among U.S. adolescents: Evidence from the 2017–2021 Youth Risk Behavior Survey. PLOS One, 21(10), e0359065. <https://doi.org/10.1371/journal.pone.0359065>
2. **Cardozo, F.**, Scoppetta, O., García-Mejía, N., & Baral, A. (2026). The declining centrality of cannabis in networks of lifetime substance use among Colombian University students, 2012-2023. International Journal of Drug Policy, 157, 105485. <https://doi.org/10.1016/j.drugpo.2026.105485>
3. **Cardozo, F.**, Brown, E. C., Mejía-Trujillo, J., Balise, R., Cañizares, C., Pérez-Gómez, A., St George, S. M., & Gabbay, V. (2026). Automating Motivational Interviewing Coding in Adolescent Substance Use Prevention: Human-AI Agreement Study. JMIR AI, 5, e95964–e95964. <https://doi.org/10.2196/95964>
4. Jaramillo, J., **Cardozo, F.**, Maya, D.H.A. et al. Towards a Better Understanding of HIV Prevention Knowledge Among Latino MSM: A Latent Class Analysis Study. AIDS Behav (2026). <https://doi.org/10.1007/s10461-026-05163-w>
5. Morales, V., **Cardozo, F.**, Balise, R., George, S., Feaster, D. (2026). The Utility of Machine Learning-Enhanced Developmental Cascade Models in Prevention Science. Prevention Science (2026). <https://doi.org/10.1007/s11121-026-01897-0>

6. Gabbay, V., Hills, C., **Cardozo, F.**, Chance, AB., Sharma, A., Mowrey, W., Baker, Z., Qi, Q., Burgess, HJ., Berman, JW., Gange, SJ., Gustafson, DR., Weber, KM., French, AL. (2026). The kynurenine pathway in relation to depression and anhedonia symptom severity among women with HIV. Brain, Behavior & Immunity - Health. 2026 Feb 16; 53:101199. doi: 10.1016/j.bbih.2026.101199. PMID: 41757375; PMCID: PMC12934288.
7. Scoppetta, O., **Cardozo, F.**, Brown, E. C., & Morales, V. (2025). Evaluating the relationship between marijuana use, aggressive behaviors, and victimization: an epidemiological study in Colombian adolescents. International journal of adolescent medicine and health, 37(1), 1–9. <https://doi.org/10.1515/ijamh-2024-0167>.
8. **Cardozo, F.**, Brown, E. C., Mejía-Trujillo, J., & Pérez-Gómez, A. (2023). Moderated mediation of motivation and self-efficacy in a brief motivational interviewing intervention to reduce youth alcohol use. Drugs: Education, Prevention and Policy, 0(0), 1–7. <https://doi.org/10.1080/09687637.2023.2167649>
9. Brown, E., Montero-Zamora, P., **Cardozo, F.**, Reyes, M., Briney, J., Mejía-Trujillo, J., & Gómez, A. (2021). A comparison of cut points for measuring risk factors for adolescent substance use and antisocial behaviors in the U.S. and Colombia. International Journal of Environmental Research and Public Health, 18, 470. <https://doi.org/10.3390/ijerph18020470>
10. Eisenberg, N., Brown, E., Mez, A., Mejía-Trujillo, J., Paredes-Aguilar, M., **Cardozo, F.**, Maipo, S., & Guttmanova, K. (2020). Community utilization of risk and protective factor data for prevention planning in Chile and Colombia. Health Promotion International, 2020, 1--13. <https://doi.org/10.1093/heapro/daaa075>
11. Reyes, M., Pinto-Gómez, C., **Cardozo, F.**, Pérez-Gómez, A., Mejía-Trujillo, J., & Toro, J. (2020). Evaluation of the prevention program brief intervention based on motivational interviewing in Colombian adolescents. International Journal of Mental Health and Addiction, 18, 1--11. <https://doi.org/10.1007/s11469-019-0057-3>
12. Pérez-Gómez, A., **Cardozo, F.**, & Salcedo-Tocarruncho, L. (2020). Association between psychoactive substance abuse and health disorders in Colombian problem drug users: Second study. International Journal of Mental Health and Addiction, 18. <https://doi.org/10.1007/s11469-018-9881-0>
13. Montero-Zamora, P., Reyes, M., **Cardozo, F.**, Brown, E., Gómez, A., Mejía-Trujillo, J., Toro, J., & Paredes, M. (2020). Adolescent substance use and its association with risk and protective factors. An exploratory analysis of the large-scale school survey of Comunidades que se Cuidan, Colombia. Adicciones, 32, 105--115. <https://doi.org/10.20882/adicciones.1083>.
14. Pérez-Gómez, A., Lanziano, C., Reyes, M., Mejía-Trujillo, J., & **Cardozo, F.** (2018). Perfiles asociados al consumo de alcohol en adolescentes colombianos. Acta Colombiana de Psicología, 21, 258--281. <https://doi.org/10.14718/ACP.2018.21.2.12>
15. Reyes, M., Mejía-Trujillo, J., Gómez, A., **Cardozo, F.**, Pinto, C., & Pinto Gómez, J. C. (2018). Effectiveness of a brief intervention based on motivational interviewing in Colombian adolescents. Psicología: Teoría e Pesquisa, 33. <https://doi.org/10.1590/0102.3772e33421>

Journal Articles (under review)

1. **Cardozo, F.**, Brown, Eric C., Mejía-Trujillo, J., Balise, R. St. George, Sara., Pérez-Gómez, A., Gabbay, V. (2026). Simulating Adolescents with Generative AI to Support Facilitator Training in a Motivational Interviewing-Based Prevention Program. Prevention Science.
2. **Cardozo, F.**, Mehra, L., Pondikos, M., Brown, E.; Delgado, G.; Zayde, A; Gustavo, D., Rubinstein, T., Alpert, J., & Gabbay, V. (2026). Parental Early-Life and Recent Adversities as Correlates of Child Emotional and Behavioral Outcomes. Journal of Psychiatric Research.

Books and Book chapters

1. **Cardozo, F.**, & Molano, A. (2016). Regresión Lineal. In L. García & J. Escobar (Eds.), Análisis estadísticos. Elementos fundamentales de análisis estadísticos en SPSS (pp. 101--108). Editorial Universidad El Bosque.

Comments

1. (Under review) **Cardozo, F.**, Mejía-Trujillo, J, Pérez-Gómez, A. (2026). Beyond Funding Refusal: Protecting Alcohol Research Where Alternatives Are Scarce. Journal of Studies on Alcohol and Drugs.

Reports and Technical Publications

1. Seminario, E., Ferrandiz Salazar, J. A., Madanes Quintanilla, P., González-Beltrán, O. L., Moreno Romero, C., Ramirez-Varela, L., **Cardozo, F.**, et al. (2024). Strengthening Monitoring and Evaluation Systems for Holistic Learning in Contexts of Emergency and Protracted Crisis: Colombia and Peru-Summary Diagnostic Report for Peru. figshare. <https://doi.org/10.6084/m9.figshare.26023423>.

2. Pérez-Gómez, A., Mejía-Trujillo, J., Reyes-Rodríguez, M. F., & **Cardozo, F.** (2015). Consumo de alcohol en menores de 18 años en Colombia: 2015. Segundo estudio en siete capitales y dos municipios pequeños. (p. 204). Dupligráficas Ltda. <https://nuevosumbos.org/website/post?id=547>

Conference Presentations

1. **Cardozo, F.** (2026). Using Generative Artificial Intelligence to Enhance Implementation Fidelity in Prevention Program Training. <https://spr.confex.com/spr/spr2026/meetingapp.cgi/Paper/36203>
2. **Cardozo, F.** (2025). Computing Inter-rater reliability for continuous scales using R. Latin American Conference on R for Research and Development. <https://latinr.org/>.
3. **Cardozo, F.** & Scoppetta, O. (2025). What Is Changing in Drug Use? An Analysis of Substance Use in Higher Education. <https://congresopsicologiacolombia.com/>.
4. **Cardozo, F.** & Cañizares, C. (2025). An Introduction to Using R for Programming in Biostatistics and Data Science, in Spanish. R/Medicine 2025 conf. https://rconsortium.github.io/RMedicine_website/.
5. **Cardozo, F.** Scoppetta, O., & Cañizares, C. (2025). From Periphery to Center? Marijuana's Changing Role in the Substance Use Network. Sunbelt 2025. International Network for Social Network Analysis. <https://www.insna.org/events/sunbelt-2025--paris>.
6. **Cardozo, F.** (2025). R Packages for Social Network. II Chilean Social Network Conference 2025. <https://www.chisocnet.org/chisocnet-conference/conference-2025>.
7. **Cardozo, F.** (2024). BreakNbuild - Tools to evaluate sample size effects on Machine Learning Models. LatinR Conference. <https://latinr.org/cronograma/programa/>
8. **Cardozo, F.**, Eric C. Brown., Scoppetta, O. (2024). Evaluating the Impact of Marijuana on Adolescent Aggression: A Population-Based Study in Colombia. Fifth Annual Graduate and Postdoctoral Research Symposium. https://events.miami.edu/event/research_symposium_2024
9. **Cardozo, F.** (2023). Exploring the interrelationships between NIH-funded projects on adolescent alcohol use: A two-mode network analysis. Sunbelt 2023. International Network for Social Network Analysis. <https://www.insna.org/events/sunbelt-2023>
10. **Cardozo, F.** (2023). Exploring the similarities in parent-child perceptions of drug use risk: The role of genetics and environment. Fifth Annual Graduate and Postdoctoral Research Symposium. https://events.miami.edu/event/research_symposium_2023
11. Pinilla, V., **Cardozo, F.**, Montero-Zamora, P., Brown, E.C. (2022). Businesses that Care: A Toolkit for Implementing Underage Drinking Prevention Programs in Communities. Oral presentation at the Social Marketing Conference, Clearwater Beach, FL.
12. Balise, R., Nahodyl, L., Calderon, A., & **Cardozo, F.** (2022). How to create publication ready tables in R. Presented at UseR conference. <https://user2022.r-project.org/program/tutorials/>
13. Brown, E.C., Oliveira Corrêa, A., **Cardozo, F.**, Guidi-Gentil, C., Montero-Zamora, P., Wells, A. (2022). A Systematic Review of Preventive Interventions for Youth Alcohol, Tobacco, and Other Drug Use in Latin America and the Caribbean. Oral presentation at the Annual Meeting of the Brazilian Society for Prevention Research, online.
14. **Cardozo, F.**, & Molano, A. (2021). Ecological factors in friendship relationships and alcohol use in scholars. Presented at Networks 2021: A Joint Conference of Sunbelt and NetSci. <https://doi.org/10.13140/RG.2.2.17630.28487>

Posters (presenter)

1. **Cardozo, F.**, Ivanova, Miglena., Cañizares, Catalina, Arrington, Myles. (2025) Mapping Adolescent Socio-Emotional Well-Being: A Network Analysis of Future of Families and Child Well-being Study. SRCD 2025 Biennial Meeting.
2. **Cardozo, F.**, & Brown, C. (2024). Data Science and Machine Learning Methodology Dissemination in Latin America. Frost Institute for Data Science and Computing.
3. **Cardozo, F.** (2024). Polydrug Use Patterns Among Colombian Adolescents: A Graph-Based Exploration. International Network for Social Network Analysis. <https://sunbelt2024.com>
4. **Cardozo, F.**, & Brown, C. (2023). Data Science and Machine Learning Methodology Dissemination in Latin America. Frost Institute for Data Science and Computing.
5. **Cardozo, F.**, Brown, C., Mejía-Trujillo, J., & Pérez-Gómez, A. (2023). Parent-Youth Correspondence in Perceptions of Risk Factors for Adolescent Drug Use: A Dyadic Analysis. Poster presented at Society for Prevention Research Annual Meeting, SPR. DOI: [10.13140/RG.2.2.25548.59521](https://doi.org/10.13140/RG.2.2.25548.59521)
6. **Cardozo, F.**, Montero-Zamora, P., Brown, C., Ortiz-García, J., & Beaulieu, D. (2023) Cost-Effectiveness Analysis

of the Tomando Buenas Decisiones Program to Reduce Family Risk Factors Associated with Alcohol Use. Poster presented at Society for Prevention Research Annual Meeting, SPR. DOI: [10.13140/RG.2.2.28904.03848](https://doi.org/10.13140/RG.2.2.28904.03848)

7. **Cardozo, F.**, Brown, E., Mejía-Trujillo, J., & Pérez-Gómez, A. (2021). Moderated mediation of motivation and self-efficacy in a brief motivational interviewing intervention to reduce youth alcohol use. Poster presented at Society for Prevention Research Annual Meeting, SPR. <https://doi.org/10.13140/RG.2.2.17630.28487>
8. **Cardozo, F.**, & Molano, A. (2019). Close friends vs. The norm effect: Effects of proximal and distal alcohol social context. Poster presented at Society for Prevention Research Annual Meeting, SPR. <https://doi.org/10.13140/RG.2.2.11232.23048>
9. **Cardozo, F.**, Guerrero, P., & Molano, A. (2017). Time-variant effects of friendship change on academic achievement: An exploratory social network analysis in Colombia. Poster presented at Society for Research in Child Development, SRCD. <https://doi.org/10.13140/RG.2.2.17630.28487>

Posters and presentations (coauthor)

1. Serban, F., **Cardozo, F.**, & Brown, E. C. (2026). Application of Generative Artificial Intelligence in Teaching Prevention Science. <https://spr.confex.com/spr/spr2026/meetingapp.cgi/Paper/36201>
2. Andris, T., **Cardozo, F.**, Nguyen, T., Mcleod, L., Gemelli, M., Vasconcello, H., Xu, J., Seifert, A., Ely, B., & Gabbay, V. (2026) *Neurometabolic signatures of adolescent depression: A longitudinal analysis linking striatal GSH to clinical symptom ratings*. Social and Affective Neuroscience Society (SANS) Annual Conference, San Diego, CA. (Poster Spotlight Selection).
3. Mcleod, L., **Cardozo, F.**, Gemelli, M., Andris, T., Vasconcello, H., Sharma, A., Starrels, J., Ely, B., & Gabbay, V. *Pain Processing and Neurocircuitry in People Living With and Without HIV*. (2026), Social and Affective Neuroscience Society (SANS) Annual Conference, San Diego, CA.
4. Serban, F., **Cardozo, F.**, & Brown, E. C. (2026). Application of Generative Artificial Intelligence in Teaching Prevention Science. Seven Annual Graduate and Postdoctoral Research Symposium. <https://www.grad.miami.edu/about/grad-events-and-calendar/research-symposium/index.html>.
5. Serban, F., **Cardozo, F.**, & Brown, E. C. (2025). Development of a generative artificial intelligence-based approach to teaching prevention science. European Society for Prevention Research Annual Meeting, Berlin, Germany.
6. Guidi, C., **Cardozo, F.**, Thurow, C., Ribeiro-Schneider, D., Regina-Noto, A., & Brown, E. (2025). Measurement Invariance by Sex in Family Risk Factors for Adolescent Alcohol and Drug Use in Brazil. Poster session presented at the annual meeting of the Society for Prevention Research, Seattle, W.
7. Pondikos, M., **Cardozo, F.**, Mehra, L., Delgado, G., Brown, E., Zayde, A., Derella, O., Rubinstein, T., Alpert, J., Gabbay, V., (2025). Parental Early-Life and Recent Adversities as Correlates of Child Emotional and Behavioral Outcomes. Presented at PRIME Research Day, Montefiore Medical Center / Albert Einstein College of Medicine.
8. Delgado, G., Gallo, L., Bhambhani, Y., **Cardozo, F.**, Holzman, S., Musicaro Yuksel, R., Hribar, B., Perez, H., & Gabbay, V. (2025). Predictors of engagement in a randomized controlled trial investigating opioid use disorder and chronic pain treatment. Presented at PRIME Research Day, Montefiore Medical Center / Albert Einstein College of Medicine.
9. Guidi, C., **Cardozo, F.**, & Brown, E. (2024). Analysis of the Brazilian Mortality National Data System (SIM) for Youth Suicide Prevention. Poster session presented at the annual meeting of the Society for Prevention Research, Washington, D.C.
10. Guidi, C., **Cardozo, F.**, & Brown, E. (2024). Risk Factors for Youth Substance Use in Brazil: A Multigroup Confirmatory Factor Analysis Comparing Age and Gender Groups. Poster session presented at the annual meeting of the Society for Prevention Research, Washington, D.C.
11. Guidi, C., **Cardozo, F.**, Ribeiro-Schneider, D., & Brown, E. (2022). Communities That Care – Brazil: Collection of Youth Risk and Protective Factor Data in Florianopolis. Poster presented at Society for Prevention Research, , Washington, D.C.
12. Montero-Zamora, P., Reyes-Rodríguez, M.F., **Cardozo, F.**, Brown, E.C., Pérez-Gómez, A., Mejía-Trujillo, J., Toro, J., Paredes-Aguilar, M. (2018). Concurrent validity of adolescent risk and protective factors for substance use in Colombia. Poster presentation at the Annual Meeting of the Society for Prevention Research, Washington, DC. Honorable mention.

Symposium Discussant

1. Perry, E. W., Walsh, C., Fortson, K., Self-Brown, S., Massetti, G. (2026). The Power of Patterns: Person-Centered Approaches for Real-World Prevention Challenges. **F. Cardozo** (Discussant), Awarded Abstract of Distinction.

Acknowledged Contributions

1. Liu, Q., Nguyen, T. N. B., Tobe, R. H., Stern, E. R., Ely, B. A., & Gabbay, V. (2026). *Neural response to reward uncertainty in adolescents with mood and anxiety symptoms*. *Neuropsychopharmacology*, 51(9), 1623-1633. <https://doi.org/10.1038/s41386-026-02412-3>
2. Panellas, I. O. (2026). *Multilevel predictors of COVID-19 mortality in Florida nursing homes and integrated resident- and facility-level analysis* [**Doctoral dissertation**, Florida International University].
3. Ivanova, M. Y. (2025). *Joint patterns, changes in joint patterns, and patterns of joint change: Examining heterogeneity in codevelopment of internalizing and externalizing problems through different developmental lenses* [**Doctoral dissertation**, The Pennsylvania State University].
4. Cañizares-Escobar, C. (2025). *Comparing supervised machine learning classification methods to identify risk factors for suicide morbidity among USA high school students* [**Doctoral dissertation**, Florida International University].
5. Forero Niño, K. (2024). *Sociomoral reasoning of non-compliance with the law in a context of illegality, violence, and differentiated presence of the State* [**Master's thesis**, Universidad Nacional de Colombia]. Institutional Repository of the Universidad Nacional de Colombia. <https://repositorio.unal.edu.co/handle/unal/86339>
6. Chirino, M. (2023). *Examining the relationship between demographics and exhibiting autism spectrum disorder behaviors with the Q-CHAT-10 screening tool* [**Master's thesis**, University of Miami].
7. Álvarez Barreto, M. A. (2022). *Understandings of peace among residents of Bogotá: Sociomoral reasoning regarding direct and structural violence* [**Master's thesis**, Universidad Nacional de Colombia].

Teaching

Instructor

1. 2026 Fall. EPH 701. Innovations in Prevention Science: Use of Large Language Models (LLMs) for Public Health Research. University of Miami. USA.
2. 2026 Spring. EPH 752: Advanced Research Methods. Public Health. University of Miami. USA.
3. 2015 – 2018. PSI 122: Quantitative Methods III. Department of Psychology, Universidad de El Bosque. Colombia.
4. 2015 - 2018 PSY 131: Program evaluation. Department of Psychology, Universidad de El Bosque. Colombia.

Teaching Assistantships

1. 2024 Spring. EPH 752: Advanced Research Methods. Department of Public Health Sciences, University of Miami.
2. 2023 June. R Medicine 101: Introduction to Clinical Data Analysis with R. R-consortium.
3. 2023 Summer. BST 692: Data Science and Machine Learning for Health Research. Department of Public Health Sciences, University of Miami.
4. 2023 Spring. BST 623: Software Tools for Manuscript Development and Reproducible Research. Department of Public Health Sciences, University of Miami.
5. 2022 Spring EPH 752: Advanced Research Methods. Department of Public Health Sciences, University of Miami.
6. 2022 Summer BST 692: Data Science and Machine Learning for Health Research. Department of Public Health Sciences, University of Miami.
7. 2021 Fall BST 623: Software Tools for Manuscript Development and Reproducible Research. Department of Public Health Sciences, University of Miami.
8. 2021 Spring. EPH 652: Research Methods. Department of Public Health Sciences, University of Miami.
9. 2016 – 2018. EDUC 4026: Advanced Research Methods. Department of Education, Universidad de Los Andes. Colombia.

Guest Lecturer

1. 2025 CISO 4604: Program Evaluation: Causal Inference. Universidad de Los Andes. Colombia.
2. 2024 EPH 752: Advanced Research Methods: Software for advanced Methods. University of Miami. USA.
3. 2024 CISO 4604: Program Evaluation Causal Inference. Universidad de Los Andes. Colombia.
4. 2023 CBPC 1069: Education in Colombia: What is prevention science. University of Los Andes. Colombia.
5. 2023 CBPC 1066: Social Policy Experiments: Causal Inference in Public Health. University of Los Andes.

Colombia.

6. 2023 CISO 4604: Project Evaluation: Causal Inference. Universidad de Los Andes. Colombia. Colombia.
7. 2022 EPH 752: Advanced Research Methods: Software Integration for Data Analysis. University of Miami. USA.
8. 2022 EPH 752: Advanced Research Methods: Dyadic Data Analysis. University of Miami. USA.
9. 2022 EPH 752: Advanced Research Methods: Monte Carlo Simulation to Estimate Power. University of Miami.
10. 2022 EPH 701: Research Methods: Causal Inference in Prevention Science. University of Miami. USA.
11. 2019 EDUC 4026: Advanced Quantitative Methods: Moderation. University of Los Andes. Colombia.
12. 2017 EPH 752: Advanced Research Methods: Social Network Analysis. University of Miami. USA.
13. 2015 EPH 752: Advanced Research Methods: Confirmatory Factor Analysis. University of Miami. USA.

Invited Talk

1. (scheduled) 2026-09-17 Harnessing Large Language Models for Reproducible and Controlled Data Extraction. University of Ilorin. Nigeria.
2. 2025-03-12 Fostering Open Science in Latin America- rOpenSci.

Workshops Designed and Delivered

1. 2026-05 Promover la equidad científica: una introducción a R. R/Medicine 2026 conf.
https://rconsortium.github.io/RMedicine_website/workshops.html#equidad.
2. 2026-01 Machine Learning for Social Sciences. Universidad de los Andes.
3. 2025-11 Introduction to the Tidymodels ecosystem for Machine Learning. Latin American Conference on R for Research and Development. Preconference Workshop.
4. 2024-05 Programming with Python. Universidad de Los Andes. Colombia. Funded by: University of Miami. Software Carpentry.
5. 2024-03 R for Reproducible Research. University of Miami. USA. Organized by: University of Miami. Software Carpentry.
6. 2024-02 Machine Learning for Classification Problems. Fundación Universitaria Konrad Lorenz. Colombia. Funded by: Frost Institute for Data Science and Computing.
7. 2024-02 Programming with Python. University of Miami. USA. Funded by: University of Miami - Software Carpentry.
8. 2024-02 Machine Learning Modeling with the R Tidyverse Ecosystem. Florida International University. USA. Organized by: Florida International University.
9. 2024-01 Mastering Effective Data Visualization with ggplot2 in R: A Hands-on Workshop for Social Work Researchers. USA. Presented in The Society for Social Work and Research Conference.
10. 2023-12 Imbalanced Data in Machine Learning Modeling. University of Miami. USA. Funded by: Frost Institute for Data Science and Computing.
11. 2023-11 Introduction to Machine Learning. International Seminar for the Evaluation of the Quality of Education (ICFES). Colombia. Funded by: Frost Institute for Data Science and Computing.
12. 2023-11 Creating Tables in R. Universidad Nacional. Colombia. Funded by: Frost Institute for Data Science and Computing.
13. 2023-05 Regression Techniques for Classification Problems. Universidad de los Andes. Colombia. Funded by: Frost Institute for Data Science and Computing.

Memberships

1. 2026 Royal Statistical Society.
2. 2025 CIMIENTOS: Multidisciplinary research collective dedicated to innovation in early childhood development and social transformation in Colombia and Latin America.
3. 2025 R-Medicine. Member of the director's board.
4. 2025 Society for Research in Child Development SRCD.
5. 2023 rOpenSci. Transforming science through open data, software & reproducibility.
6. 2023 Software Carpentry. Teaching researchers the computing skills.
7. 2022 Society for Prevention Research SPR.
8. 2021 International Network for Social Network Analysis INSNA.

Professional Service

1. 2026 Reviewer, Addiction Science & Clinical Practice.

2. 2026 Reviewer, BMC Public Health.
3. 2026 Reviewer, Journal of Adolescent Health.
4. 2025 Reviewer, Prevention Science.
5. 2025 Reviewer, Journal of Behavioral Medicine.

Programming

R Packages

1. 2026 - ghosted: De-identify transcript batches with one local app.
2. 2025 - DedooseR: Monitoring and Analyzing Dedoose Qualitative Data Exports.
3. 2024 - wandR: Making R learning engaging through gamified tools.
4. 2023 - BreakNBuild: Evaluate ML model performance under progressive data splits.
5. 2023 - rUM: Templates for reproducible workflows at the University of Miami.
6. 2022 - DocumentData: Tools to document datasets effectively.
7. 2022 - ChessOlympiad: Datasets for multilevel modeling practice.

Interactive web applications

1. 2024 - SNA Explorer: Visualization tool for network package dependencies and collaboration.
2. 2023 - Creating Pretty Tables in R: Interactive tutorial, University of Miami.
3. 2022 - Bias Simulation: Shiny app to illustrate reporting bias in drug studies.
4. 2022 - Reshape Data Tutorial: Long/wide transformations in R.
5. 2022 - Underage Drinking Prevention Toolkit: Public health resource site.

Skills

1. Advance level in statistical software: Mplus, Stata, Excel, SAS, SPSS.
2. Qualitative Software: Nvivo, Dedoose.
3. Advance level in programming languages: R, Python, Julia, JavaScript.
4. Reproducible Report: Quarto, Markdown, R Shiny Apps, Jupyter.
5. DevOps: Git, GitHub, Docker.
6. Web development: React, Node.js, CSS, HTML, GitHub Pages, Netlify.

Research Projects

Current

Investigating the Neural and Clinical Effects of Cannabis Use in Adolescents with Depression

This NIH-funded longitudinal study examines how cannabis use may influence brain development and clinical outcomes among adolescents with depression. The project focuses on reward and pain neurocircuits, including the habenula, a small brain structure implicated in mood regulation, motivation, and responses to pain and loss. Using advanced neuroimaging, clinical assessments, and biological measures, the study evaluates whether cannabis use and depression have additive effects that contribute to altered brain functioning, greater symptom persistence, and poorer mental health outcomes over a two-year follow-up period. The study enrolls adolescents across a wide range of depression severity and cannabis use patterns, alongside healthy controls, and combines functional MRI tasks, resting-state imaging, repeated psychiatric evaluations, daily cannabis use surveys, and laboratory testing of cannabinoid exposure.

Funding Source: NIH 5R01DA059527-02

Role: Postdoctoral Fellow - Data Scientist

Evaluating a Multimodal Parenting Intervention to Address COVID-19 Impact in the Bronx

This NIH-funded project evaluates the psychological and healthcare access impact of a parenting intervention for vulnerable caregivers in the Bronx following the COVID-19 pandemic. The study uses a randomized controlled trial to assess the effects of a 12-session Mentalization-Based Therapy (CARE) program and a HIPAA-compliant mobile health app (Valera) on stress, healthcare access, and reflective functioning among caregivers of children with psychiatric or autoimmune conditions, and frontline healthcare workers. As the project's Data Scientist, I developed and maintained workflows for data extraction from REDCap via API, implemented reproducible pipelines for instrument scoring and validation, and led the preparation of analysis-ready datasets. I also collaborated on study design proposals, advised on statistical methodology, and contributed to manuscript development. My role emphasizes the integration of data science into the research cycle, enabling efficient data use, reproducible computation, and high-impact insights for improving mental health in underserved populations.

Funding Source: NIH PAR-20-237
Role: Postdoctoral Fellow - Data Scientist

2026 OpenAI Researcher Access Program - Research support in the form of API credits

Memory-enabled conversational AI systems can store sensitive user information, but they may also create new sensitive information through inference. This study asks whether memory-enabled chatbots infer sensitive substance-use behavior from prior conversations, even when the user never disclosed it. We ask: (1) Do memory-enabled chatbots infer that a user has used recreational drugs based on conversational history? (2) Is this inference more likely when the user previously discussed drugs, prevention, treatment, or harm reduction, without disclosing personal use? (3) Do chatbots endorse a sensitive claim about the user more readily indirectly (embedded in a list) than when asked directly? We call this "personalized hallucination": false, unsupported, or inferred claims about a user generated from conversational memory and personalization signals.

Role: Principal Investigator

Amount: \$ 1.000 USD API credits.

Past

2024-2025 Development of an LLM-Based chatbot for Program Implementation Fidelity

I led the development of an AI-powered system to support implementation fidelity for the IBEM (Brief Intervention in Motivational Interviewing) program. This work involved designing and evaluating a Large Language Model based chatbot to simulate realistic participant interactions, as well as creating an automated feedback platform that provides facilitators with tailored, actionable insights on their adherence to the program steps. The system draws on data from facilitator–chatbot interactions to generate fidelity reports, refined through mixed-methods research including expert feedback, certified implementer input, and practitioner surveys and focus groups. By integrating motivational interviewing principles with scalable AI technologies, this project demonstrates how automated tools can enhance training, improve fidelity, and expand the reach of evidence-based preventive interventions.

Role: Principal Investigator

2023 Expanding the Use of Collaborative Data Science at UM: Machine Learning Methodology Dissemination in Latin America

An initiative funded by the Institute for Data Science and Computing; this project aimed to strengthen collaborative data science networks across Latin America. Through targeted workshops on machine learning training and resource dissemination, I outreach to multiple institutions to foster data-driven research and capacity-building throughout the region.

Funding Source: Institute for Data Science and Computing.

Grant: # CD230007

Amount: \$ 20,000.00 USD.

Role: Co-PI

2021 Organization of American States (OAS) - Inter-American Drug Abuse Control Commission (CICAD) and the University of Miami (UM): First Hemispheric Drug Use Prevention Forum

Contributed to the design, collection, and organization of a comprehensive dataset on drug use prevention interventions across Latin America and the Caribbean, supporting regional efforts to understand and address substance use challenges.

Funding Source: Copeland Foundation Award (DPHS Internal Application).

Grant #GR0153355 (Award # CFA02-2020).

Role: Graduate Research Assistant.

Languages

Spanish, English